

AutoMate — AI, Speech & Adaptive Logic Track

Role confirmation for team sign-off

Field	Detail
Prepared by	Muhammad Hamza Habib
Track	D3 — AI, Speech & Adaptive Logic
Project	AutoMate — AI-assisted mobile application supporting autistic children
Event	Alibaba Cloud AI Hackathon Pakistan 2026 — theme: AI for Pakistan's Future
Build phase	22–27 August 2026 (regional round 28–30 August, Grand Finale 10 September)
Date issued	20 August 2026
Status	Awaiting team confirmation — please respond by end of 21 August

1. Purpose of this document

I am taking ownership of the AI, speech and adaptive-logic track (D3) for the AutoMate build. This document sets out exactly what I will build, the features that work will produce in the app, and the technology I intend to use, so that the team can confirm it before the build phase opens on 22 August.

Please read Section 6 in particular. Two items there change what other tracks build, so they need an answer before Day 0 (21 August) rather than during the sprint.

2. Features my work will produce in the app

These are the things a user will actually see and hear because of this track. Everything marked P0 is committed; the single P1 item is a stretch goal that will only be attempted once all P0 work is complete.

#	Feature	What the user experiences	Priority
1	Bilingual speech throughout the app	Every card, button, activity prompt and routine step can be spoken aloud in English or Urdu. Works with no internet connection.	P0
2	Tap-to-talk sentence building (AAC)	The child taps picture cards and the phone speaks a complete sentence — "I want an apple." This is the flagship feature of the app.	P0
3	Genuinely correct Urdu	Urdu is generated with its own sentence structure rather than translated word-by-word, and the verb agrees with the child's gender. The same taps produce correct sentences in both languages.	P0
4	A grid that learns the child	Cards the child uses often move to the front of the grid automatically, with no setup required from the parent.	P0
5	The emotion recognition game	A face is shown, the child picks the emotion, gets instant feedback, and receives a score and stars at the end of the session.	P0
6	Difficulty that adapts	The activity becomes harder after repeated success and easier after repeated mistakes. The wrong answers themselves get harder to tell apart, not just more numerous. A parent can lock the level.	P0

7	Spoken daily routines	Tapping a step in the child's daily schedule reads it aloud in the chosen language.	P0
8	"Can you make a happy face?"	A live camera activity where a meter fills as the child smiles. Runs entirely on the phone — no photo or video is ever saved or uploaded.	P1 — stretch

In short: if the app speaks, builds a sentence, scores an activity, adapts its difficulty, or reads a face — that is my track.

3. What I will build (technical deliverables)

#	Component	Purpose	Est.
A1	TtsService	Bilingual text-to-speech: queued, interruptible, warm-started so the first word starts in under half a second, and fully offline.	6 h
A2	SentenceRealiser	Turns a strip of tapped cards into a grammatical sentence — English word order and articles, Urdu word order and verb agreement.	6 h
A3	CardRanker	Ranks cards by recency-weighted usage so the grid personalises itself to the child.	2 h
A4	EmotionActivityEngine	Generates questions, selects wrong answers by visual confusability, scores the session and awards stars.	6 h
A5	AdaptiveLevelController	The rule-based promotion and demotion logic between support levels, with a parent override.	3 h
A6	Routine narration	Speaks routine steps and transition warnings through the speech service.	2 h
A7	ExpressionPracticeService	On-device face detection driving the smile-practice activity. P1 — stretch goal only.	8 h

Committed P0 work totals roughly 25 hours across Days 1–5. A7 is not committed and will only begin if A1–A6 are complete and no one is blocked on me.

4. Technology stack I will use

Layer	Technology	Why this one
Speech output	flutter_tts (platform text-to-speech)	The only maintained Flutter wrapper around the device's own speech engine. Runs offline, costs nothing, and supports both English and Urdu where the device has the voice.
On-device ML (P1)	google_mlkit_face_detection	Ships a working on-device model, roughly 30–60 ms per frame, no network calls. Gives us smile and eye-open probabilities directly.
Camera (P1)	camera	The official Flutter camera plugin. Its image stream is what feeds the face detector.
Exposing my services	flutter_riverpod	Team standard, already chosen by the data track. I add no state-management opinion of my own.
All language and activity logic	Plain Dart — no libraries	Sentence generation, ranking, scoring and adaptive rules are all written by hand. No ML runtime, no NLP library, and no code generation (build_runner is a known time sink in a short

		sprint).
Testing	flutter_test	Every algorithm in this track is pure Dart, so it can be unit tested without a device or a database. I expect to carry most of the team's test coverage.
Development environment	Qoder (issued 21 August)	The hackathon's AI IDE. I will use it for the mechanical parts of the track — test scaffolding, model classes, camera plumbing — and keep concrete examples, as judges are likely to ask how it was used.

4.1 Two stack decisions that need the team

1. audioplayers — conditional, decided on 21 August.

On 21 August I will test whether our demo phone actually has an Urdu voice installed. If it does not, we fall back to pre-recorded Urdu audio for the core vocabulary, which needs one extra package because flutter_tts cannot play audio files. This is the single largest risk in my track and I am retiring it before the build phase opens rather than discovering it mid-sprint.

2. Alibaba Cloud Model Studio (Qwen) — optional, decided on Day 5.

This is an Alibaba Cloud hackathon, so it is worth deciding deliberately whether we use one of their AI services rather than omitting it by accident. The only safe place for a generative model in this product is the parent-facing weekly summary — turning aggregated numbers into two or three plain sentences. It would never face the child, and would send aggregate counts only. It requires a Firebase Blaze upgrade to do without hard-coding an API key. My recommendation: decide this on Day 5, only if all P0 work is green, and never put it on the critical path.

4.2 What I am deliberately not using, and why

Each of these is a plausible question from a judge, so the reasoning is worth agreeing as a team now.

Rejected	Reason
A custom-trained emotion classifier (TFLite)	Three to four days of uncertain work with a real chance of ending the sprint with nothing running. Google's on-device face detection gives us a better live demo in about four hours. Documented honestly as future work.
Cloud speech synthesis	Breaks our offline requirement, adds latency and cost, and would send a child's utterances to a server. The device's own engine is free, instant and local.
Open-ended AI chat for the child	Explicitly out of scope in our own scope document on child-safety grounds. It cannot be adequately guardrailed at this scale.
A machine-learned adaptive difficulty policy	There is no ethically obtainable training data for this, and a parent needs to understand why the level changed. Rule-based is the correct choice here, not a compromise.
A translation API for Urdu	It would produce wrong word order exactly where it matters most. Generating each language from a shared structure is both cheaper and better.

5. What I do not own

Stated so that no one duplicates work or edits the same files. My code lives in the services and logic folders and never touches the interface, the database or the translation files.

Not mine	Whose it is
Any screen, widget or theme	D1 — Frontend & Design System
Firestore reads and writes, data models, security rules	D2 — Firebase & Data
Wiring my services into screens	D4 — Integration & Release
Translation files and user-facing text	D1 — I emit only keys and structured data, never English strings
The dashboard, charts and progress aggregation	D4 — I produce the session result object they aggregate

6. What I need from the team

These are the things that block or weaken my track if they do not happen. Item 1 and item 5 are the urgent ones.

#	What I need	From	By
1	Four extra fields on every seeded communication card: part of speech, countable or not, whether the English label starts with a vowel sound, and the Urdu grammatical gender. Without these the Urdu sentence generation degrades to simple word-joining and we lose the strongest moment in the demo. It is four fields, seeded once.	D2	Day 1
2	The child profile needs to carry gender. Urdu verbs agree with the speaker, so this is what lets the app say the correct form for a girl and for a boy.	D2 / D1	Day 1
3	Confirmation of the session-result object shape (score, total, level played, level after, duration, stars). This object crosses three tracks, so changing it mid-sprint is expensive.	D4	Day 1
4	Six emotion face images — happy, sad, angry, surprised, scared, neutral. Illustrations rather than photographs, which avoids any likeness or consent issue and matches our sensory-friendly design.	D1	Day 3
5	Confirmation of which physical Android device is the demo device, and access to it on 21 August so I can verify the Urdu voice. Emulators are not reliable for this. A second device with the same voice would remove a single point of failure.	Team	Day 0
6	A decision on the optional Alibaba Cloud summary described in Section 4.1.	Team	Day 5

7. My delivery schedule

Mapped onto the real hackathon dates. Note that the build phase gives us six days, not seven — there is no spare day at the end, so rehearsal happens on the evening of the 27th.

Date	Day	What I deliver	Gate
Thu 21 Aug	Day 0	Environment set up; Urdu voice verified on the physical demo device.	Urdu risk retired
Fri 22 Aug	Day 1	Speech service working in both languages on hardware. Interface signatures published to the team.	The app speaks
Sat 23 Aug	Day 2	All sentence, scoring, ranking and adaptive logic written and unit tested. Interfaces frozen.	30+ tests green
Sun 24 Aug	Day 3	The full AAC pipeline: card tap to sentence to speech, in both languages, offline.	Flagship works
Mon 25 Aug	Day 4	Emotion activity end to end, scoring into the database, difficulty	Session records

		adapting.	
Tue 26 Aug	Day 5	Routine narration. All committed work complete. P1 begins only if everything else is green.	P0 complete
Wed 27 Aug	Day 6	Hardening, offline and Urdu audit, rehearsal. Code freeze at midday.	Demo-ready

8. Interface freeze

I will publish the exact function signatures the rest of you code against on the evening of Day 1, marked provisional, and re-publish them on the evening of Day 2 marked frozen. After that point a change costs the whole team time and needs everyone's agreement. This is deliberate: it means D1, D2 and D4 can write code against my track before my track is finished, so nobody sits waiting on me.

9. Confirmation

By signing below you are confirming that this is the work you expect me to own, that the technology choices in Section 4 are agreed, and that the requests in Section 6 will be met on the dates given.

Name	Track	Confirmed	Date
Muhammad Hamza Habib	D3 — AI, Speech & Adaptive Logic	Yes	20 Aug 2026
	D1 — Frontend & Design System		
	D2 — Firebase & Data		
	D4 — Integration & Release		
	Supervisor / Mentor		

Full technical detail — algorithms, interface signatures, test plan and risk register — is in `AutiMate_D3_AI_Track_Plan.md` in the project repository.